Project 2 : Toxic Comment Classification

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1) From Project 1 to Project 2

- **Best BoW representation:** uni+bigrams (1,2), shape = (159571, 2905614)
- **Best classical ML model: SVM model**, **fast to train**, offers good **interpretability**, and performs well with **BoW/TF-IDF representations**, but it does not capture the **semantic context** of language, which limits its effectiveness compared to **contemporary approaches**.

2) Word embeddings

With **MiniLM embeddings** (384 dimensions), **encoding** is slower but **training** is fast, making **deployment** feasible. **Logistic Regression** is more practical than **SVM** because it outputs **probabilities** and allows **threshold tuning**, giving a trade-off between **precision** and **recall**. Unlike **BoW**, which needed **heavy preprocessing**, **MiniLM** only requires **light cleaning** and shows weaker results with **linear models**, which is expected because its full potential appears with **modern architectures** like fine-tuned **Transformers**.

Table 1: Comparison of linear models using Bag-of-Words (BoW) vs MiniLM sentence embeddings.

Model	**Toxic F1**	**Macro F1**	**ROC-AUC**	**PR-AUC (toxic)**
BoW + Logistic Regression	0.74	0.86	0.972	0.856
BoW + Linear SVM	0.80	0.89	0.976	0.880
MiniLM + Logistic Regression	0.64	0.79	0.9625	0.8096
MiniLM + Linear SVM	0.65	0.80	0.9633	0.8139

3) Transformers

Model used: RoBERTa-base, an optimized version of **BERT**, with improved **training procedure** (more **data** and **iterations**, removal of the **Next Sentence Prediction** task).

```
=== RoBERTa-base (weighted + smoothing) — TEST ===
              precision    recall  f1-score   support

   non-toxic      0.984     0.983     0.983     28856
       toxic      0.839     0.848     0.843      3059

    accuracy                          0.970     31915
   macro avg      0.911     0.916     0.913     31915
weighted avg      0.970     0.970     0.970     31915

ROC-AUC: 0.9881 | PR-AUC (toxic): 0.9243
Train time (s): 61430.5
```

Figure 1: RoBERTa-base Classification Report

It achieves the best performance compared to **DistilBERT** used in project 1, with **ROC-AUC = 0.9881** vs 0.9864.

4) Prompt engineering

We used the **FLAN-T5** model, a sequence-to-sequence (encoder-decoder) architecture. It is flexible handling many NLP tasks, but more computationally expensive than RoBERTa (which only uses an encoder). FLAN-T5 is very sensitive to the exact wording of prompts and may shift in its predictions, whereas GPT model are more robust since they were trained on more diverse contexts.

3 prompt experiments (Macro-F1) :

Prompt type	Macro-F1
Zero-shot	0.93
Role prompting	0.93
Few-shot	0.09

Prompt-based approaches remain more costly than RoBERTa and less direct for classification (require generating output tokens rather than explicit probabilities).

5) Evaluation

Figure 2: Classification report for each model

Model	Non-toxic Precision	Non-toxic Recall	Non-toxic F1-score	Toxic Precision	Toxic Recall	Toxic F1-score	Macro-F1	ROC-AUC	PR-AUC(toxic)
RoBERTa-base(weighted)	0.984	0.983	0.983	0.839	0.848	0.843	0.913	0.9881	0.9243
Baseline TF-IDF(1-2g)+ LinearSVC	0.979	0.979	0.979	0.800	0.799	0.800	0.889	0.9776	0.8841
MiniLM + LinearSVC	0.987	0.909	0.946	0.507	0.888	0.646	0.796	0.9633	0.8139
MiniLM + Logistic Regression	0.987	0.908	0.946	0.504	0.887	0.643	0.794	0.9625	0.8096
Zero-shot	0.9838	0.9897	0.9868	0.8975	0.8466	0.8713	0.9290		
Few-shot	0.0000	0.0000	0.0000	0.0958	1.0000	0.1749	0.0875		
Role prompting	0.9842	0.9886	0.9864	0.8881	0.8499	0.8686	0.9275		

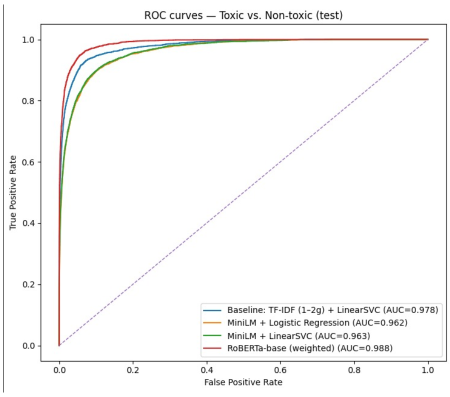

Figure 3: ROC Curves for different models

Best performing approach : **RoBERTa-base** and **Zero-shot prompt**

6) Reflexion

While recent approaches achieve strong performance, their adoption in practice requires careful reflection on additional aspects including computational cost, potential biases, transparency, and privacy.

Table 2: Comparison of key aspects across different approaches

Aspect	Key points
Cost	• **Embeddings**: ∼3–4min, scalable, good trade-off time/quality. • **Transformers**: ∼2h training (GPU), inference slower → better for batch processing. • **LLMs**: ∼6–7h training, ∼1h30 classification per prompt. • **Deployment**: choice depends on balance between computational cost and performance.
Bias	• Biases come from training data. • Ex: if "woman" often appears with insults, the model may wrongly classify "She is a woman" as toxic. • Biases are not intentional but reflect the statistical nature of data. • Risk: reduced reliability, unjustified censorship, ethical concerns.
Transparency	• **Classical models (BoW, LR)**: easiest to explain → each word has a weight, easy to show influence. • **Transformers (RoBERTa)**: harder to explain (black-box) → contextual embeddings and multi-layer attention. • **Prompting with LLMs**: hardest → results depend on both internal representations and prompt wording.
Privacy	• User comments are sent to an external platform for storage and analysis. • **Risks for Wikipedia**: exposure to legal constraints (e.g., GDPR), loss of confidentiality, reduced control/sovereignty, and potential decline in user trust.

Conclusion: Classical models remain **simple** and **fast**, but recent approaches (**Transformers**, **LLMs**) provide **contextual understanding** of language and significantly higher **performance**. The choice of model therefore depends on the trade-off between **performance**, **computational cost**, **explainability**, and **deployment constraints**.
[1]

[1] we used AI/tool assistance for the code and Hugging Face website for the transformer model